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Smart Drone-Based Facade Cleaning

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Reinventing Facade Cleaning Through Smart Drone Technology

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Industry Problem

Facade cleaning is labor-intensive and high-risk, heavily dependent on rope-access workers. Current systems are inefficient, water-heavy, lack integration with building maintenance intelligence, and struggle with increasing ESG reporting pressure.

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The Core Insight

"Touchless spray drones" are ineffective for heavy urban dirt. Effective cleaning requires physical contact. Our concept is a "Flying Floor Scrubber" with an active rotary brush, integrating thermal-based maintenance audits during the cleaning process.

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Why Now?

Stricter EU environmental laws and increasing manual labor costs have made traditional cleaning methods risky and expensive. Simultaneously, the facility management industry is actively seeking automated, compliant solutions, creating an urgent market demand for our technology.

From Cleaning Service to Data-Driven Infrastructure Solution

Smart Drone-Based Facade Cleaning for Commercial Buildings (DACH)

What We Do

- Drone-based façade cleaning using active rotary brush.
- Targeted micro-cleaning with significantly lower water usage.
- Simultaneous thermal imaging during cleaning.
- Automated maintenance reporting.
- Generates ESG-ready façade condition reports aligned with EU sustainability reporting requirements

Value Created

- Reduces liability exposure for Facility Management (FM) contractors
- Reduces water and chemical consumption.
- Improves cleaning effectiveness in urban environments.
- Converts cleaning into actionable building intelligence.
- Enables proactive maintenance decisions.

Initial market: Commercial buildings in Germany → scalable across DACH.

(Disruption) vs (Value Chain): Structural Trade-offs

Customer Access	Value Chain (through FM incumbents)	Disruption (to asset owners / FM managers)
Revenue Model	Hardware-as-a-Service (HaaS) + recurring service fees	Full-service margin capture
Speed to Market	Leverage FM firms’ existing contracts (Faster)	Must acquire first 3–5 anchor clients (Slower)
Capital Requirement	Moderate (R&D + compliance + limited sales team)	High (sales + brand + operations)
Learning Type	Learn integration complexity inside FM workflows	Learn end-customer switching behavior & pricing sensitivity
Brand Positioning	Technology supplier	Competing cleaning provider
Risk Exposure	Dependence on partner adoption rate	Direct liability exposure

For our venture, early pilots indicate shorter sales cycles and lower resistance when embedded within FM workflows. Therefore, capital efficiency and speed of validation favor the Value Chain strategy.

Why Can’t We Do Both?

Different Organizational Capabilities Required

Value Chain requires FM integration and compliance expertise.
Disruption requires direct sales, fleet management, and brand building.

Different Sales Channels and Brand Positioning

Selling through FM incumbents positions us as an enabling technology partner.
Selling directly to asset owners positions us as a competing cleaning provider, requiring different messaging, pricing logic, and market approach.

Conflicting Incentives (Partner vs. Competitor)

If we pursue direct cleaning contracts, FM firms may perceive us as a competitive threat and resist adopting our technology.
This channel conflict would undermine trust and slow integration into existing service contracts.

Resource Dilution in Early-Stage Venture

Simultaneously investing in advanced drone R&D, regulatory compliance, direct sales, brand building, and fleet scaling would overextend capital and management capacity at an early stage.

Regulatory & Operational Focus Cannot Be Split

Operating under EU drone regulations requires concentrated compliance effort and operational discipline.
Dividing focus between hardware innovation and direct cleaning operations increases execution and liability risk.

> Early-stage ventures must concentrate scarce resources on one coherent strategic logic.

Learning Differences

Value Chain Strategy Teaches Us:

- True operational cost per façade
- Integration complexity within existing FM contracts
- Whether ESG documentation drives partner adoption
- Partner switching costs

Disruption Strategy Teaches Us:

- Direct end-user willingness to switch providers
- Price sensitivity and margin potential
- Sales cycle length
- Full-margin unit economics

> The learning paths differ: one validates integration efficiency, the other tests independent demand.

Test to Choose Between Strategies

Experiment Design:

- Run one paid pilot through FM partner (Strategy A path)
- Simultaneously approach 5 direct asset owners for pilot proposals (Strategy B path)

Compare:

Metric	FM Partner Pilot	Direct Asset Owner Pilot
Sales Cycle Length	< 4 weeks (via existing contracts)	> 8–12 weeks (new approval process)
Gross Margin Potential	Moderate (shared value chain)	High (full-service capture)
Adoption Resistance	Low	Moderate
Customer Acquisition Cost	Moderate	High (direct sales effort)
Operational Friction	Moderate (integration effort)	High

Success Criteria:

- Shorter and more predictable sales cycle
- Attractive margin after real operational costs
- Lower adoption resistance
- Higher probability of repeat contracts

Overall, the preferred strategy should demonstrate superior capital efficiency, faster validation, and lower ecosystem friction.

Does Choosing One Create Irreversible Costs?

Value Chain → Pivot to Disruption:

- Loss of neutrality with FM service partners
- Need to build direct sales and cleaning operations
- Higher marketing and liability exposure

Disruption → Pivot to Value Chain:

- Trust barrier with FM service incumbents
- Perceived competitive threat
- Repositioning required from competitor to supplier

Value Chain preserves strategic optionality more effectively in early stages.

Decision Framework

We choose based on:

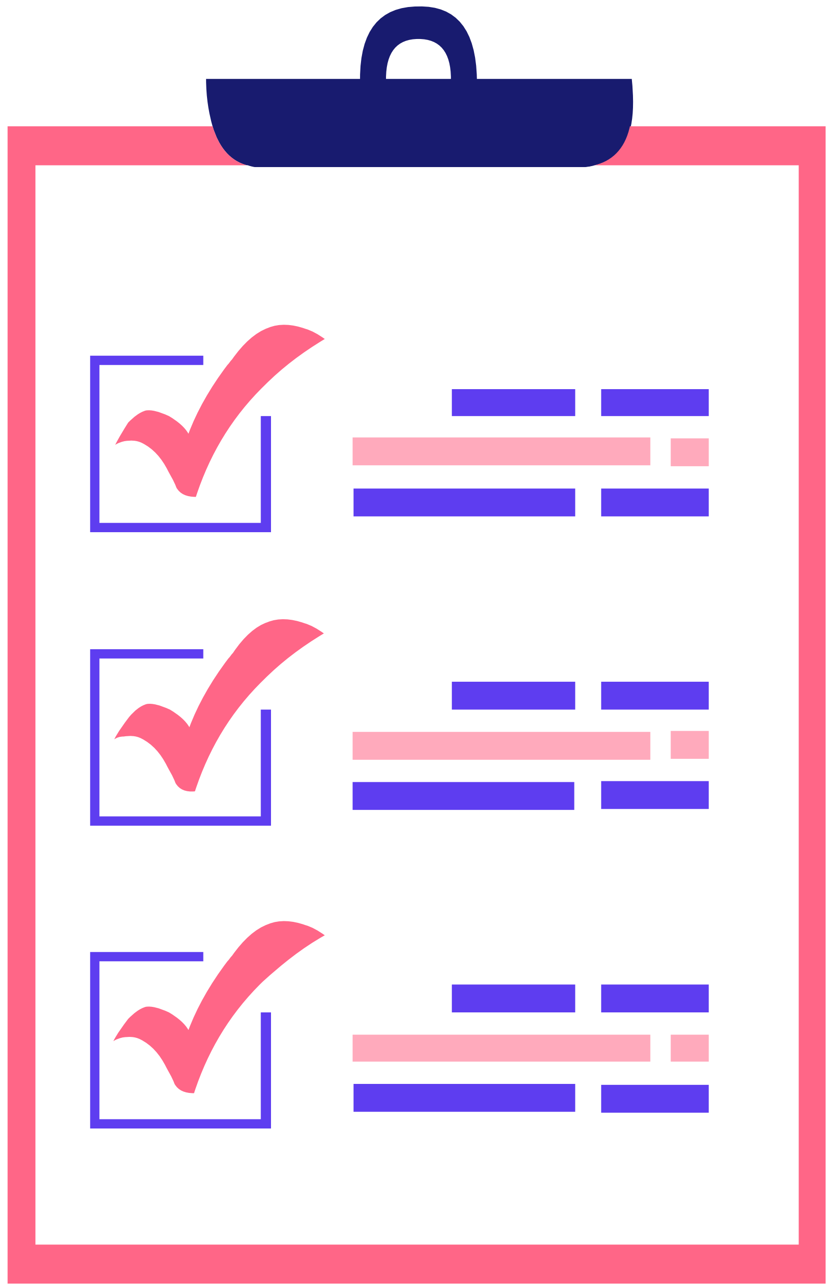
1 **Alignment with Founding Team Capabilities**
The preferred strategy demonstrates superior capital efficiency, faster validation, and lower ecosystem friction.

2 **Capital Efficiency**
Lower upfront capital by leveraging partner contracts instead of building a direct cleaning workforce.

3 **Speed of Validation**
Faster pilot deployment through FM partners compared to acquiring independent asset owners.

4 **Regulatory Feasibility**
Focused compliance under EU drone regulations without simultaneous operational scaling complexity.

5 **Strategic Optionality**
Preserves the ability to move toward selective vertical integration once credibility and data are established.



Chosen Path: Value Chain Strategy

Decision Rationale:

- Lower ecosystem friction through FM service integration
- Faster validation via existing service contracts
- Strong alignment with EU regulatory constraints
- Capital-efficient scaling model
- Builds operational credibility within the industry
- Preserves future vertical integration option (Disruption)

Disruption remains a Phase 2 strategic evolution.

Strategy C – Hybrid Evolution Path

- Start by embedding technology as a supplier within the existing value chain to minimize risk and speed up market entry.
- Establish operational dominance and collect proprietary data.
- Build brand credibility during the foundational phase.
- Gradually introduce direct premium services to selected clients.
- Transition from a cooperative role to strategic vertical integration over time.

Approach: Collaborate first to reduce risk → integrate selectively once a defensible market position is achieved.